Complex Analysis: Contours, Fourier Transforms, and Schwarz–Christoffel

Problems with background and complete solutions

Branches of $\log f$ on simply connected domains — with spaced layout

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Problem

Let U be a simply connected domain in the complex plane and let f be holomorphic on U with $f(z) \neq 0$ for all $z \in U$.

(1) Show that there exists a holomorphic F on U with $e^F = f$ (a branch of $\log f$ on U).

(2) Give an example showing that a holomorphic branch of the *scalar* logarithm need not exist on the image $f(U)$.

Background and tools

(i) Logarithms via primitives.

If f is holomorphic and never zero on a domain D, then $\dfrac{f'}{f}$ is holomorphic on D. If there exists a holomorphic F with $F' = \dfrac{f'}{f}$, then

$$\frac{(e^F)'}{e^F} = F' = \frac{f'}{f} \qquad \Longrightarrow \qquad \left(\log \frac{e^F}{f}\right)' = 0,$$

so e^F/f is constant; adding a constant to F we obtain $e^F = f$.

(ii) Primitives on simply connected domains.

If g is holomorphic on a simply connected domain U, then g has a global primitive G on U; equivalently,

$$\oint_\gamma g(z)\,dz = 0 \quad \text{for every closed loop } \gamma \subset U.$$

(iii) Integral and winding criteria.

For any closed loop γ in $\{f \neq 0\}$,

$$\oint_\gamma \frac{f'(z)}{f(z)}\,dz \;=\; 2\pi i \cdot \mathrm{Wind}\big(f\circ\gamma,\, 0\big).$$

Thus a holomorphic branch of $\log f$ on a domain $D \subset \{f \neq 0\}$ exists iff the integral above is 0 for all closed loops in D (equivalently, $f \circ \gamma$ never winds around 0).

(iv) Scalar $\log$ **on a planar domain.**

A holomorphic branch of the scalar logarithm on a domain $V \subset \mathbf{C}\setminus\{0\}$ exists iff V is simply connected (i.e. V does not wrap around the origin). Annuli about 0 and $\mathbf{C}\setminus\{0\}$ itself fail this.

Solution to (1): Existence on U

Because f is holomorphic and nonvanishing, the function

$$g(z) := \frac{f'(z)}{f(z)}$$

is holomorphic on U. Since U is simply connected, there exists a holomorphic primitive G with $G' = g$ on U.

Define

$$H(z) := e^{G(z)}.$$

Then

$$\frac{H'(z)}{H(z)} = G'(z) = \frac{f'(z)}{f(z)} \qquad \Longrightarrow \qquad \Big(\log\frac{H(z)}{f(z)}\Big)' = 0.$$

Hence H/f is constant on U. Choose $c \in \mathbf{C}$ with $e^c = f(z_0)/H(z_0)$ for some (hence every) $z_0 \in U$, and set

$$F(z) := G(z) + c.$$

Then $e^{F(z)} = f(z)$ on U, so F is a holomorphic branch of $\log f$.

Uniqueness up to $2\pi i\mathbf{Z}$. If F_1 and F_2 both satisfy $e^F = f$, then $e^{F_1 - F_2} \equiv 1$, so $F_1 - F_2$ is a constant of the form $2\pi i\, k$ with $k \in \mathbf{Z}$.

Equivalent criteria on U

Integral criterion. A holomorphic F with $e^F = f$ exists on U if and only if

$$\oint_\gamma \frac{f'(z)}{f(z)}\,dz = 0 \quad \text{for every closed loop } \gamma \subset U.$$

Winding criterion. Equivalently, a holomorphic branch of $\log f$ exists on U iff

$$\mathrm{Wind}\big(f \circ \gamma, 0\big) = 0 \quad \text{for every closed loop } \gamma \subset U.$$

Solution to (2): Examples where a scalar $\log$ on $f(U)$ fails

Global example. Let $U = \mathbf{C}$ and $f(z) = e^z$. Then f is entire, never zero, and a branch $\log f$ exists on U (take $F(z) = z$). However

$$f(U) = \mathbf{C} \setminus \{0\},$$

which is not simply connected. For the unit circle $\Gamma = \{|w| = 1\}$,

$$\oint_\Gamma \frac{dw}{w} = 2\pi i \neq 0.$$

If a holomorphic scalar logarithm existed on $\mathbf{C} \setminus \{0\}$, its derivative would be $1/w$, forcing this integral to be 0 for all closed loops—a contradiction.

Bounded variant (annulus image). Let $U = \{z : \ |z| < 2\}$ and $f(z) = e^z$. Then $\log f$ exists on U (again $F(z) = z$), but

$$f(U) = \{e^{x+iy} : \ |x| < 2, \ |y| < 2\} = \{w : \ e^{-2} < |w| < e^2\},$$

which is an annulus encircling 0. Any circle $\{|w| = r\}$ with $e^{-2} < r < e^2$ satisfies

$$\oint \frac{dw}{w} = 2\pi i \neq 0,$$

so no holomorphic scalar logarithm exists on $f(U)$.

Geometric/topological intuition

The exponential map $\exp : \mathbf{C} \to \mathbf{C} \setminus \{0\}$ is a covering map with deck transformations $z \mapsto z + 2\pi i k$ ($k \in \mathbf{Z}$). A global holomorphic scalar logarithm on a target domain $V \subset \mathbf{C} \setminus \{0\}$ would be a single-valued holomorphic inverse on V; this exists exactly when every loop in V lifts to a closed loop, i.e. when V is simply connected. Annuli and $\mathbf{C} \setminus \{0\}$ are not, so monodromy around 0 prevents a single-valued holomorphic logarithm there.

For $\log f$ on U, the only obstruction would be that some image loop $f(\gamma)$ winds around 0. On simply connected U with $f \neq 0$, the form f'/f is exact, so all such windings vanish and a branch exists.

Summary

On any simply connected domain U, every holomorphic nonvanishing f admits a holomorphic F with $e^F = f$ (unique up to $2\pi i\,\mathbf{Z}$). Nevertheless, a holomorphic scalar logarithm on $f(U)$ can fail to exist when $f(U)$ winds around the origin (e.g. $f = \exp$ gives $f(U) = \mathbf{C} \setminus \{0\}$ or an annulus).

Example. Let $U = \mathbb{C}$ and $f(z) = e^z$.

Existence of $\log f$ *on* U*.* Since f is entire and never zero, $f'/f \equiv 1$ is entire. Because U is simply connected, $G(z) := z$ is a primitive of f'/f. Define $F(z) := G(z) = z$. Then $e^{F(z)} = e^z = f(z)$, so F is a holomorphic branch of $\log f$ on U. Uniqueness holds up to $2\pi i\mathbb{Z}$: if F_1, F_2 satisfy $e^{F_j} = f$, then $e^{F_1 - F_2} \equiv 1$, hence $F_1 - F_2 \in 2\pi i\mathbb{Z}$ is a constant.

Failure of a branch of $\log$ *on* $f(U)$*.* We have $f(U) = \mathbb{C}^\times$. If there were a holomorphic branch L of the scalar logarithm on $\mathbb{C}^\times$, then $L'(w) = 1/w$ and, for every closed loop $\Gamma \subset \mathbb{C}^\times$,

$$\oint_\Gamma \frac{dw}{w} = \oint_\Gamma L'(w)\,dw = 0.$$

Taking $\Gamma = \{\, |w| = 1 \,\}$ yields $\oint_{|w|=1} \frac{dw}{w} = 2\pi i \neq 0$, a contradiction. Therefore, no holomorphic branch of the scalar log exists on $\mathbb{C}^\times$.

Monodromy/winding viewpoint. For any closed loop Γ in a domain that avoids 0,

$$\oint_\Gamma \frac{dw}{w} = 2\pi i \cdot \mathrm{Wind}(\Gamma, 0).$$

Thus a global holomorphic branch of log on a domain $V \subset \mathbb{C}^\times$ exists iff every loop in V has winding number 0 about 0 (equivalently, V is simply connected). Here $V = \mathbb{C}^\times$ has nontrivial loops around 0, so monodromy obstructs single-valuedness.

Analytic continuation around a circle (explicit jump). Start with a local branch of log near $w = 1$ and analytically continue once along the unit circle $w = e^{it}$, $t \in [0, 2\pi]$. The continuation returns to the starting point but accumulates the increment

$$\int_0^{2\pi} \frac{d}{dt}\big(\log(e^{it})\big)\,dt \;=\; \int_0^{2\pi} i\,dt \;=\; 2\pi i,$$

so the value jumps by $2\pi i$. This nontrivial monodromy forbids a single-valued holomorphic logarithm on $\mathbb{C}^\times$.

Bounded variant (annulus image). Let $U = \{z : |z| < 2\}$ and $f(z) = e^z$. Then $F(z) = z$ gives a branch of $\log f$ on U, but

$$f(U) = \{e^{x+iy} \;:\; |x| < 2,\ |y| < 2\} = \{w \in \mathbb{C}^\times : e^{-2} < |w| < e^2\},$$

an annulus encircling 0. Any circle $\{|w| = r\}$ with $e^{-2} < r < e^2$ satisfies $\oint \frac{dw}{w} = 2\pi i \neq 0$, so no holomorphic branch of the scalar log exists on $f(U)$.

Local vs. global behavior. On any simply connected subdomain $V \subset \mathbb{C}^\times$ that does not wrap around 0 (e.g. a slit plane or a sector of angle $< 2\pi$), one *can* choose a holomorphic branch of the scalar log. The obstruction is purely topological: encircling 0 forces a $2\pi i$ jump.

Connection with arg *and* f'/f. For $w \neq 0$, $\log w = \log|w| + i \arg w$; the impossibility of a global continuous arg on $\mathbb{C}^\times$ is the same obstruction. Equivalently, if L existed on V, then $(\log \circ f)' = (f'/f)$ would be the derivative of a *global* holomorphic function on V. On simply connected U this always holds for f'/f (hence $\log f$ exists on U); on $f(U)$ it fails exactly when some loop winds around 0.

Takeaway. A holomorphic nonvanishing f on a simply connected U always has a holomorphic logarithm F with $e^F = f$ (unique up to $2\pi i\mathbb{Z}$). But a holomorphic scalar logarithm on $f(U)$ may not exist if $f(U)$ winds around 0 (as with $f = \exp$ giving $\mathbb{C}^\times$ or an annulus).

- f is entire and never zero, so $f'/f \equiv 1$ is entire.
- On the simply connected set $U = \mathbb{C}$, a primitive of f'/f is $G(z) = z$.
- Set $F(z) := G(z) = z$. Then $e^{F(z)} = e^z = f(z)$. Thus F is a (holomorphic) branch of $\log f$ on U.

2) Why a holomorphic log cannot exist on $f(U) = \mathbb{C}^\times$

- $f(U) = \mathbb{C} \setminus \{0\}$ is not simply connected (its fundamental group is $\mathbb{Z}$); any loop going once around 0 has nonzero winding number.
- If a holomorphic branch of the scalar logarithm Log existed on $\mathbb{C}^\times$, then $(\mathrm{Log})'(w) = 1/w$ and hence, for every closed loop $\Gamma \subset \mathbb{C}^\times$,

  $$\oint_\Gamma \frac{dw}{w} \;=\; \oint_\Gamma (\mathrm{Log})'(w)\,dw \;=\; 0.$$

  But for the unit circle $\{|w| = 1\}$,

  $$\oint_{|w|=1} \frac{dw}{w} \;=\; 2\pi i \;\neq\; 0.$$

  Contradiction. Therefore no holomorphic branch of log exists on $\mathbb{C}^\times$.

3) Geometry/topology intuition (why the obstruction appears)

- $\exp : \mathbb{C} \to \mathbb{C}^\times$ is a *universal covering map.* Adding $2\pi i$ in z–space leaves e^z unchanged: $e^{z+2\pi i} = e^z$. So loops around 0 in $\mathbb{C}^\times$ lift to paths in $\mathbb{C}$ whose endpoints differ by $2\pi i$.

- A single-valued holomorphic log on $\mathbb{C}^\times$ would be a global holomorphic inverse to exp; that would force all lifted loops to close, which they do not—this is the *monodromy* obstruction.
- Equivalent "metric picture": horizontal lines $y = c$ map to rays of angle c; vertical lines $x = c$ map to circles $|w| = e^c$. The image wraps endlessly around 0; choosing a continuous argument globally is impossible.

Variant (same phenomenon). Take $U = \{\, |z| < 2 \,\}$ and $f(z) = e^z$. Then

$$f(U) = \{\, e^{x+iy} : \ |x| < 2, \ |y| < 2 \,\} = \{\, w \in \mathbb{C}^\times : \ e^{-2} < |w| < e^2 \,\},$$

an annulus around 0. The same integral test around any circle in that annulus, $\oint \frac{dw}{w} = 2\pi i$, blocks a global holomorphic log there.

Cauchy integral with principal value and boundary limits

Problem (restated)

Let Γ be a simple smooth positively oriented closed curve in the plane, and let f be continuously differentiable on Γ. Define, for $z \in \mathbb{C} \setminus \Gamma$,

$$F(z) := \frac{1}{2\pi i} \int_\Gamma \frac{f(\zeta)}{\zeta - z}\, d\zeta.$$

This integral diverges when z approaches a point $\zeta_0 \in \Gamma$ with $f(\zeta_0) \neq 0$. Define the principal value at ζ_0 by

$$\mathrm{P.V.}\ \frac{1}{2\pi i} \int_\Gamma \frac{f(\zeta)}{\zeta - \zeta_0}\, d\zeta \ := \ \lim_{\varepsilon \to 0^+} \frac{1}{2\pi i} \int_{\Gamma_\varepsilon} \frac{f(\zeta)}{\zeta - \zeta_0}\, d\zeta,$$

where Γ_ε is Γ with a small open arc of radius ε around ζ_0 removed (the remaining part is traversed with the original orientation).

Background and basic facts

Analyticity off the curve. For $z \notin \Gamma$ the integrand is continuous on Γ and the function F is well defined and holomorphic in each of the two components of $\mathbb{C} \setminus \Gamma$ (the interior and the exterior). Differentiation under the integral sign gives

$$F'(z) = \frac{1}{2\pi i} \int_\Gamma \frac{f(\zeta)}{(\zeta - z)^2}\, d\zeta, \qquad z \in \mathbb{C} \setminus \Gamma.$$

Recovery of interior analytic data. If f extends to a function holomorphic in the interior of Γ and continuous on $\overline{\mathrm{int}(\Gamma)}$, then the Cauchy integral formula yields

$$F(z) = f(z) \quad \text{for } z \text{ in the interior of } \Gamma, \qquad F(z) = 0 \quad \text{for } z \text{ in the exterior and } |z| \text{ large.}$$

Principal value meaning. At $z = \zeta_0 \in \Gamma$ the kernel $(\zeta - \zeta_0)^{-1}$ has a simple pole on the contour. The principal value extracts the finite, symmetric part of the singular integral by removing a small arc around the pole and letting its size shrink to 0.

Nontangential limits and the Sokhotski–Plemelj formulas

Let $F^{\pm}(\zeta_0)$ denote the limits of $F(z)$ as $z \to \zeta_0 \in \Gamma$ from the interior $(+)$ and from the exterior $(-)$ along nontangential approaches. If $f \in C^1(\Gamma)$, then these limits exist and

$$\boxed{\begin{aligned} F^+(\zeta_0) &= \frac{1}{2} f(\zeta_0) \; + \; \mathrm{P.V.}\ \frac{1}{2\pi i} \int_\Gamma \frac{f(\zeta)}{\zeta - \zeta_0}\, d\zeta, \\ F^-(\zeta_0) &= -\frac{1}{2} f(\zeta_0) \; + \; \mathrm{P.V.}\ \frac{1}{2\pi i} \int_\Gamma \frac{f(\zeta)}{\zeta - \zeta_0}\, d\zeta. \end{aligned}}$$

Consequently,

$$\boxed{F^+(\zeta_0) - F^-(\zeta_0) = f(\zeta_0)} \qquad \text{and} \qquad \boxed{\frac{F^+(\zeta_0) + F^-(\zeta_0)}{2} = \mathrm{P.V.}\ \frac{1}{2\pi i} \int_\Gamma \frac{f(\zeta)}{\zeta - \zeta_0}\, d\zeta.}$$

Sketch of derivation (indentation method)

Fix $\zeta_0 \in \Gamma$ and replace a small arc of Γ near ζ_0 by a circle of radius ε centered at ζ_0, lying inside (for F^+) or outside (for F^-) the domain. Then

$$F(z) = \frac{1}{2\pi i} \int_{\Gamma_\varepsilon} \frac{f(\zeta)}{\zeta - z}\, d\zeta \; + \; \frac{1}{2\pi i} \int_{C_\varepsilon} \frac{f(\zeta)}{\zeta - z}\, d\zeta,$$

where C_ε is the small circular arc. Let $z \to \zeta_0$ from the chosen side. The first integral tends to the principal value term. On C_ε, write $f(\zeta) = f(\zeta_0) + o(1)$ and compute

$$\frac{1}{2\pi i} \int_{C_\varepsilon} \frac{f(\zeta_0)}{\zeta - \zeta_0}\, d\zeta = \pm \frac{1}{2} f(\zeta_0),$$

with the sign $+$ for the inner indentation and $-$ for the outer one (by orientation). Adding the contributions yields the two Plemelj formulas above.

Remarks

- The regularity assumption $f \in C^1(\Gamma)$ can be weakened (e.g. Hölder continuity suffices for the nontangential limits).
- If Γ is a line, these formulas reduce to the classical Hilbert transform jump relations.
- Orientation matters: reversing the orientation of Γ switches the signs in the $\pm\frac{1}{2} f(\zeta_0)$ terms.

Concrete example on the unit circle. Let $\Gamma = \{\zeta : |\zeta| = 1\}$ (counterclockwise) and $f(\zeta) = \zeta^m$ with $m \in \mathbb{Z}_{\geq 0}$. Define

$$F(z) = \frac{1}{2\pi i} \int_{|\zeta|=1} \frac{\zeta^m}{\zeta - z}\, d\zeta, \qquad z \in \mathbb{C} \setminus \Gamma.$$

Inside/outside values. By Cauchy's integral formula, if $|z| < 1$ then $F(z) = z^m$, while if $|z| > 1$ then $F(z) = 0$. Thus F is analytic on each side of Γ, equals z^m in the interior, and vanishes in the exterior.

Boundary limits and principal value. For $\zeta_0 \in \Gamma$, the Sokhotski–Plemelj formulas yield

$$F^+(\zeta_0) = \frac{1}{2}\zeta_0^m + \mathrm{P.V.}\, \frac{1}{2\pi i} \int_{|\zeta|=1} \frac{\zeta^m}{\zeta - \zeta_0}\, d\zeta, \qquad F^-(\zeta_0) = -\frac{1}{2}\zeta_0^m + \mathrm{P.V.}\, \frac{1}{2\pi i} \int_{|\zeta|=1} \frac{\zeta^m}{\zeta - \zeta_0}\, d\zeta.$$

Since $F^+(\zeta_0) = \zeta_0^m$ and $F^-(\zeta_0) = 0$, it follows that

$$\boxed{\mathrm{P.V.}\, \frac{1}{2\pi i} \int_{|\zeta|=1} \frac{\zeta^m}{\zeta - \zeta_0}\, d\zeta = \frac{1}{2}\, \zeta_0^m}, \qquad \boxed{F^+(\zeta_0) - F^-(\zeta_0) = \zeta_0^m = f(\zeta_0)}.$$

Special case $m = 0$. For $f(\zeta) \equiv 1$ we have $F(z) = 1$ for $|z| < 1$ and $F(z) = 0$ for $|z| > 1$, and on Γ,

$$F^+(\zeta_0) = 1, \qquad F^-(\zeta_0) = 0, \qquad \mathrm{P.V.}\, \frac{1}{2\pi i} \int_{|\zeta|=1} \frac{1}{\zeta - \zeta_0}\, d\zeta = \frac{1}{2}.$$

Inside/outside values (unit circle $\Gamma = \{|\zeta| = 1\}$). Consider

$$F(z) = \frac{1}{2\pi i} \int_{|\zeta|=1} \frac{\zeta^m}{\zeta - z}\, d\zeta, \qquad m \in \mathbb{Z}_{\geq 0}.$$

Case $|z| < 1$ (pole inside). As a function of ζ, the integrand has a simple pole at $\zeta = z$ lying inside Γ. By Cauchy's integral formula,

$$\frac{1}{2\pi i} \int_{|\zeta|=1} \frac{\zeta^m}{\zeta - z}\, d\zeta = \zeta^m\big|_{\zeta=z} = z^m.$$

Equivalently, $\mathrm{Res}_{\zeta=z} \frac{\zeta^m}{\zeta - z} = z^m$, so the integral is $2\pi i\, z^m$, and dividing by $2\pi i$ yields $F(z) = z^m$.

Case $|z| > 1$ (no pole inside). Now the pole $\zeta = z$ lies outside Γ, hence the integrand is holomorphic on and inside the contour. By Cauchy's theorem,

$$\frac{1}{2\pi i} \int_{|\zeta|=1} \frac{\zeta^m}{\zeta - z}\, d\zeta = 0, \quad \text{i.e.} \quad F(z) = 0.$$

Series check for $|z| > 1$ (optional). For $|z| > 1$,

$$\frac{\zeta^m}{\zeta - z} = -\frac{1}{z}\, \frac{\zeta^m}{1 - \zeta/z} = -\frac{1}{z} \sum_{k=0}^{\infty} \frac{\zeta^{m+k}}{z^k} \qquad \big(|\zeta/z| < 1\big).$$

Integrating termwise over $|\zeta| = 1$ gives $\oint \zeta^n\, d\zeta = 0$ for all $n \neq -1$, so the whole integral is 0, confirming $F(z) = 0$ outside.

Cauchy principal value on a smooth curve: existence and formula

Problem (exact restatement of the two tasks)

(a) Show that the principal value is well defined for all $\zeta_0 \in \Gamma^*$, namely that the limit

$$\mathrm{P.V.}\ \frac{1}{2\pi i}\int_{\Gamma} \frac{f(\zeta)}{\zeta-\zeta_0}\,d\zeta \;:=\; \lim_{\varepsilon\downarrow 0} \frac{1}{2\pi i}\int_{\Gamma_\varepsilon} \frac{f(\zeta)}{\zeta-\zeta_0}\,d\zeta$$

exists, where Γ is a simple smooth positively oriented curve, f is continuously differentiable on Γ^*, and Γ_ε is the part of Γ that is at least ε–away from ζ_0.

(b) Show that

$$\mathrm{P.V.}\ \frac{1}{2\pi i}\int_{\Gamma} \frac{f(\zeta)}{\zeta-\zeta_0}\,d\zeta \;=\; \frac{1}{2\pi i}\int_{\Gamma} \frac{f(\zeta)-f(\zeta_0)}{\zeta-\zeta_0}\,d\zeta \;+\; \frac{1}{2}f(\zeta_0).$$

Background and notation

Let Γ be a simple C^1 positively oriented closed curve in the plane and let $f \in C^1(\Gamma)$. For $z \in \mathbb{C}\setminus\Gamma$ define the Cauchy transform

$$F(z) := \frac{1}{2\pi i}\int_{\Gamma} \frac{f(\zeta)}{\zeta - z}\,d\zeta.$$

Then F is holomorphic on each component of $\mathbb{C}\setminus\Gamma$. As $z \to \zeta_0 \in \Gamma$, the kernel $(\zeta - z)^{-1}$ has a simple pole on the contour; the principal value removes a small arc around ζ_0 and takes the limit as its length shrinks to 0.

Solution to (a): existence of the principal value

Fix $\zeta_0 \in \Gamma$. Split the integrand into a regular part and a pure singular kernel:

$$\frac{f(\zeta)}{\zeta-\zeta_0} = \underbrace{\frac{f(\zeta)-f(\zeta_0)}{\zeta-\zeta_0}}_{\text{regular near } \zeta_0} + \underbrace{\frac{f(\zeta_0)}{\zeta-\zeta_0}}_{\text{pure kernel}}. \tag{1}$$

Regular part. Since $f \in C^1(\Gamma)$, there is a constant C with

$$|f(\zeta) - f(\zeta_0)| \le C\,|\zeta-\zeta_0| \quad \text{for } \zeta \text{ near } \zeta_0.$$

Hence the quotient $\dfrac{f(\zeta)-f(\zeta_0)}{\zeta-\zeta_0}$ is bounded and continuous on Γ (interpreted by continuity at ζ_0). Therefore

$$\int_{\Gamma_\varepsilon} \frac{f(\zeta)-f(\zeta_0)}{\zeta-\zeta_0}\,d\zeta \quad \text{converges as } \varepsilon \downarrow 0,$$

so its contribution to the principal value has a proper limit (indeed, it is just the ordinary integral over Γ).

Pure kernel. On the small missing arc of radius ε around ζ_0 the integral

$$\int \frac{d\zeta}{\zeta - \zeta_0}$$

is $\pm\,\pi i$ in the limit (half of the $2\pi i$ residue), with the sign determined by whether the indentation is taken inside or outside, together with the positive orientation of Γ. Thus the contribution of the pure kernel to the truncated integral is uniformly bounded and tends, in the principal value sense, to the finite number $\dfrac{1}{2}\,2\pi i \cdot \dfrac{f(\zeta_0)}{2\pi i} = \dfrac{1}{2} f(\zeta_0)$ (precisely accounted for in part (b) below).

Combining the two parts from (1) shows that the limit defining

$$\text{P.V. } \frac{1}{2\pi i} \int_\Gamma \frac{f(\zeta)}{\zeta - \zeta_0}\, d\zeta$$

exists for every $\zeta_0 \in \Gamma$.

Solution to (b): decomposition formula with the $\frac{1}{2} f(\zeta_0)$ term

Start from the splitting (1) and integrate over Γ_ε:

$$\frac{1}{2\pi i} \int_{\Gamma_\varepsilon} \frac{f(\zeta)}{\zeta - \zeta_0}\, d\zeta = \frac{1}{2\pi i} \int_{\Gamma_\varepsilon} \frac{f(\zeta) - f(\zeta_0)}{\zeta - \zeta_0}\, d\zeta + \frac{f(\zeta_0)}{2\pi i} \int_{\Gamma_\varepsilon} \frac{d\zeta}{\zeta - \zeta_0}.$$

Let $\varepsilon \downarrow 0$. The first term converges to the *ordinary* integral over Γ because its integrand is bounded and continuous on Γ:

$$\lim_{\varepsilon \downarrow 0} \frac{1}{2\pi i} \int_{\Gamma_\varepsilon} \frac{f(\zeta) - f(\zeta_0)}{\zeta - \zeta_0}\, d\zeta = \frac{1}{2\pi i} \int_\Gamma \frac{f(\zeta) - f(\zeta_0)}{\zeta - \zeta_0}\, d\zeta.$$

For the second term,

$$\lim_{\varepsilon \downarrow 0} \frac{1}{2\pi i} \int_{\Gamma_\varepsilon} \frac{d\zeta}{\zeta - \zeta_0} = \frac{1}{2},$$

because the missing small arc carries half of the $2\pi i$ residue at $\zeta = \zeta_0$ (with the positive orientation prescribed in the statement). Therefore,

$$\text{P.V. } \frac{1}{2\pi i} \int_\Gamma \frac{f(\zeta)}{\zeta - \zeta_0}\, d\zeta = \frac{1}{2\pi i} \int_\Gamma \frac{f(\zeta) - f(\zeta_0)}{\zeta - \zeta_0}\, d\zeta + \frac{1}{2} f(\zeta_0),$$

as required.

Concrete example on the unit circle

Let $\Gamma = \{\zeta : |\zeta| = 1\}$ (counterclockwise), $\zeta_0 = e^{it_0}$, and $f(\zeta) = \zeta^m$ with $m \geq 0$. Then

$$\mathrm{P.V.}\ \frac{1}{2\pi i} \int_{|\zeta|=1} \frac{\zeta^m}{\zeta - \zeta_0}\, d\zeta = \frac{1}{2\pi i} \int_{|\zeta|=1} \frac{\zeta^m - \zeta_0^m}{\zeta - \zeta_0}\, d\zeta + \frac{1}{2}\, \zeta_0^m.$$

But the integrand on the right is regular on the unit circle, and Cauchy's formula gives the value

$$\frac{1}{2\pi i} \int_{|\zeta|=1} \frac{\zeta^m - \zeta_0^m}{\zeta - \zeta_0}\, d\zeta = \frac{1}{2}\, \zeta_0^m,$$

so the principal value equals $\dfrac{1}{2}\zeta_0^m$. This matches the Plemelj jump relations: the interior limit is ζ_0^m and the exterior limit is 0, hence their average is exactly the principal value.

Connection between the principal value and the index of the curve

For a smooth closed curve Γ and a point ζ_0 not on Γ, the *index* (or winding number) of Γ about ζ_0 is defined by

$$\mathrm{Ind}(\Gamma; \zeta_0)\ :=\ \frac{1}{2\pi i} \oint_\Gamma \frac{d\zeta}{\zeta - \zeta_0}.$$

It counts how many times Γ winds around ζ_0, with orientation. For a positively oriented simple Jordan curve,

$$\mathrm{Ind}(\Gamma; \zeta_0) = \begin{cases} 1, & \zeta_0 \text{ inside } \Gamma, \\ 0, & \zeta_0 \text{ outside } \Gamma. \end{cases}$$

When $\zeta_0 \in \Gamma$, the integral above diverges because of the singularity at $\zeta = \zeta_0$. However, its *Cauchy principal value* exists and equals *half* the index:

$$\mathrm{P.V.}\ \frac{1}{2\pi i} \oint_\Gamma \frac{d\zeta}{\zeta - \zeta_0}\ =\ \frac{1}{2}\, \mathrm{Ind}(\Gamma; \zeta_0).$$

For a positively oriented simple curve, $\mathrm{Ind}(\Gamma; \zeta_0) = 1$, so

$$\mathrm{P.V.}\ \frac{1}{2\pi i} \oint_\Gamma \frac{d\zeta}{\zeta - \zeta_0} = \frac{1}{2}.$$

General decomposition in terms of the index

In the decomposition formula

$$\mathrm{P.V.}\ \frac{1}{2\pi i} \oint_\Gamma \frac{f(\zeta)}{\zeta - \zeta_0}\, d\zeta = \frac{1}{2\pi i} \oint_\Gamma \frac{f(\zeta) - f(\zeta_0)}{\zeta - \zeta_0}\, d\zeta + \frac{1}{2}\, f(\zeta_0),$$

the last term $\frac{1}{2} f(\zeta_0)$ comes precisely from the principal value of the pure kernel. Replacing the constant $\frac{1}{2}$ by $\frac{1}{2}\,\mathrm{Ind}(\Gamma; \zeta_0)$ gives the more general form valid for any winding number:

$$\boxed{\mathrm{P.V.}\ \frac{1}{2\pi i} \oint_\Gamma \frac{f(\zeta)}{\zeta - \zeta_0}\, d\zeta = \frac{1}{2\pi i} \oint_\Gamma \frac{f(\zeta) - f(\zeta_0)}{\zeta - \zeta_0}\, d\zeta + \frac{\mathrm{Ind}(\Gamma; \zeta_0)}{2}\, f(\zeta_0).}$$

Consequences for boundary limits (Plemelj jump)

Define

$$F(z) = \frac{1}{2\pi i} \oint_\Gamma \frac{f(\zeta)}{\zeta - z}\, d\zeta, \qquad z \in \mathbb{C} \setminus \Gamma.$$

Then the nontangential limits from the interior $(+)$ and exterior $(-)$ satisfy

$$F^\pm(\zeta_0) = \mathrm{P.V.}\ \frac{1}{2\pi i} \oint_\Gamma \frac{f(\zeta)}{\zeta - \zeta_0}\, d\zeta \ \pm\ \frac{\mathrm{Ind}(\Gamma;\zeta_0)}{2} f(\zeta_0), \qquad \zeta_0 \in \Gamma.$$

Hence the *jump relation*

$$\boxed{F^+(\zeta_0) - F^-(\zeta_0) = \mathrm{Ind}(\Gamma;\zeta_0)\, f(\zeta_0).}$$

Remarks

- For a positively oriented simple contour, $\mathrm{Ind}(\Gamma;\zeta_0) = 1$, so the jump equals $f(\zeta_0)$ and the principal value term corresponds to $\frac{1}{2} f(\zeta_0)$.
- If Γ winds k times around ζ_0, then $\mathrm{Ind}(\Gamma;\zeta_0) = k$, and all the $\frac{1}{2}$ coefficients are multiplied by k.
- When ζ_0 lies outside Γ, $\mathrm{Ind} = 0$, so the principal value is simply $\frac{1}{2\pi i} \oint_\Gamma \frac{f(\zeta)}{\zeta - \zeta_0}\, d\zeta = 0$.

(c) Interior/Exterior limits and the jump

Let $F(z) = \frac{1}{2\pi i} \int_\Gamma \frac{f(\zeta)}{\zeta - z}\, d\zeta$ with Γ a simple C^1 Jordan curve and $f \in C^1(\Gamma)$. For $\zeta_0 \in \Gamma$,

$$\lim_{\substack{z \to \zeta_0 \\ z \text{ inside } \Gamma}} F(z) = \mathrm{P.V.}\ \frac{1}{2\pi i} \int_\Gamma \frac{f(\zeta)}{\zeta - \zeta_0}\, d\zeta + \frac{1}{2} f(\zeta_0),$$

$$\lim_{\substack{z \to \zeta_0 \\ z \text{ outside } \Gamma}} F(z) = \mathrm{P.V.}\ \frac{1}{2\pi i} \int_\Gamma \frac{f(\zeta)}{\zeta - \zeta_0}\, d\zeta - \frac{1}{2} f(\zeta_0).$$

Thus F has a jump $F^+(\zeta_0) - F^-(\zeta_0) = f(\zeta_0)$.

Proof sketch. Split $\dfrac{f(\zeta)}{\zeta - \zeta_0} = \dfrac{f(\zeta) - f(\zeta_0)}{\zeta - \zeta_0} + \dfrac{f(\zeta_0)}{\zeta - \zeta_0}$. The first term yields the principal value. The second contributes $\pm\frac{1}{2} f(\zeta_0)$ via an indentation around ζ_0 (the short circular arc integral equals $\pm\pi i$).

(d) What if Γ is not simple / only piecewise smooth? What if f is not C^1?

Non-simple curves (multiple windings). Replace $\pm\frac{1}{2}$ by $\pm\frac{1}{2}\,\mathrm{Ind}(\Gamma;\zeta_0)$:

$$F^\pm(\zeta_0) = \mathrm{P.V.}\ \frac{1}{2\pi i} \int_\Gamma \frac{f(\zeta)}{\zeta - \zeta_0}\, d\zeta \ \pm\ \frac{\mathrm{Ind}(\Gamma;\zeta_0)}{2} f(\zeta_0), \qquad F^+ - F^- = \mathrm{Ind}(\Gamma;\zeta_0)\, f(\zeta_0).$$

Piecewise smooth / Lipschitz curves. For Γ Lipschitz (e.g. piecewise C^1), the same formulas hold for *nontangential* limits. The Cauchy singular integral operator is bounded on $L^p(\Gamma)$, $1 < p < \infty$, so Plemelj limits exist a.e.

Weaker regularity on f. If $f \in C^{0,\alpha}(\Gamma)$ (Hölder), the limits hold pointwise for all $\zeta_0 \in \Gamma$. If $f \in L^p(\Gamma)$, $1 < p < \infty$, the jump relations hold in the L^p (and a.e.) sense.

Example (unit circle). For $\Gamma = \{|\zeta| = 1\}$ and $f(\zeta) = \zeta^m$ $(m \geq 0)$:

$$F(z) = \begin{cases} z^m, & |z| < 1, \\ 0, & |z| > 1, \end{cases} \quad \Rightarrow \quad F^+(e^{it}) - F^-(e^{it}) = e^{imt} = f(e^{it}),$$

and $\mathrm{P.V.}\ \frac{1}{2\pi i} \int_{|\zeta|=1} \frac{\zeta^m}{\zeta - e^{it}}\, d\zeta = \frac{1}{2} e^{imt}$.

(ii) Curve only piecewise smooth / Lipschitz (corners allowed). The Plemelj formulas remain valid for *Lipschitz* curves (e.g. piecewise C^1 with finite length) when boundary limits are taken *nontangentially.* This is classical in singular–integral theory (Calderón–Zygmund): the Cauchy singular integral operator is bounded on $L^p(\Gamma)$ for $1 < p < \infty$, and the nontangential boundary limits exist almost everywhere.

(iii) Regularity of f.

- For classical *pointwise* limits at every boundary point, it suffices to assume $f \in C^{0,\alpha}(\Gamma)$ (Hölder) for some $\alpha \in (0, 1]$.
- For *almost-everywhere* (a.e.) limits and L^p jump relations, it is enough to assume $f \in L^p(\Gamma)$ with $1 < p < \infty$.
- The original hypothesis $f \in C^1(\Gamma)$ is more than enough; it guarantees the principal value is well defined and that the interior/exterior limits exist *at every* $\zeta_0 \in \Gamma$ (no exceptional set).

Cauchy integrals on the unit circle: analysis of the integrands and evaluation

Problem

Let γ be the positively oriented unit circle $\{|z| = 1\}$. Use Cauchy Integral Formulas to evaluate

$$\text{(a)} \quad \int_\gamma \frac{e^z \cos z}{z}\, dz, \qquad \text{(b)} \quad \int_\gamma \frac{e^z \cos z}{z^3}\, dz.$$

Background

- **Cauchy Integral Formula (CIF).** If f is holomorphic on and inside γ and a lies inside, then
$$\frac{1}{2\pi i}\int_\gamma \frac{f(\zeta)}{\zeta - a}\,d\zeta = f(a), \qquad \frac{1}{2\pi i}\int_\gamma \frac{f(\zeta)}{(\zeta - a)^{n+1}}\,d\zeta = \frac{f^{(n)}(a)}{n!}.$$
- **Residues.** $\displaystyle\int_\gamma g(z)\,dz = 2\pi i \sum \operatorname{Res}(g; \text{poles inside } \gamma).$
- e^z and $\cos z$ are entire; hence $N(z) := e^z \cos z$ is entire.

Anatomy of the integrands

Let $N(z) = e^z \cos z$.

- **Zeros of N.** e^z has no zeros; $\cos z$ has zeros at $z = \frac{\pi}{2} + k\pi$ for $k \in \mathbb{Z}$ (simple zeros), none on $|z| = 1$.
- **Poles.** $\dfrac{N(z)}{z}$ has a simple pole at $z = 0$. $\dfrac{N(z)}{z^3}$ has a triple pole at $z = 0$. Because $N(0) = 1 \neq 0$, the pole orders are not reduced.
- **Domains.** Both $\dfrac{N}{z}$ and $\dfrac{N}{z^3}$ are holomorphic on $\mathbb{C} \setminus \{0\}$; the only singularity inside γ is at 0.
- **Magnitude along lines.** With $z = x + iy$,
$$|N(z)| = e^x\,|\cos(x+iy)|, \qquad |\cos(x+iy)|^2 = \cos^2 x\,\cosh^2 y + \sin^2 x\,\sinh^2 y.$$
Thus along vertical lines (x fixed) $|N| \sim e^x \frac{e^{|y|}}{2}$ for large $|y|$; along horizontal lines (y fixed) $|N|$ oscillates with an e^x envelope.

Evaluations

(a) The integral of $\dfrac{N(z)}{z}$. By CIF with $f = N$ and $a = 0$,
$$\int_\gamma \frac{e^z \cos z}{z}\,dz = 2\pi i\,N(0) = 2\pi i\,e^0 \cos 0 = 2\pi i.$$

(b) The integral of $\dfrac{N(z)}{z^3}$. By the higher-order CIF with $n = 2$,
$$\int_\gamma \frac{e^z \cos z}{z^3}\,dz = 2\pi i\,\frac{N^{(2)}(0)}{2!}.$$

Compute derivatives:

$$N'(z) = e^z(\cos z - \sin z), \qquad N''(z) = e^z(\cos z - \sin z) + e^z(-\sin z - \cos z) = -2e^z \sin z,$$

so $N''(0) = 0$. Therefore

$$\int_\gamma \frac{e^z \cos z}{z^3}\, dz = 0.$$

Remarks

- Laurent expansions at 0 show directly that the coefficient of $1/z$ in $\dfrac{N}{z^3}$ is 0; hence the integral vanishes.
- The jump/limit issues do not arise here since the integrands are holomorphic on and inside γ except at 0, which is isolated and handled by CIF.

Detailed anatomy of $N(z) = e^z \cos z$

Algebraic forms.

$$N(z) = e^z \cos z = \tfrac{1}{2}\Big(e^{(1+i)z} + e^{(1-i)z}\Big).$$

Thus N is entire of order 1 and exponential type $\sqrt{2}$.

Zeros and critical points.

- Zeros: $e^z \neq 0$, so $N(z) = 0 \iff \cos z = 0 \iff z = \frac{\pi}{2} + k\pi$ for $k \in \mathbb{Z}$ (all simple).
- Critical points: $N'(z) = e^z(\cos z - \sin z) = 0 \iff \tan z = 1 \iff z = \frac{\pi}{4} + k\pi$ (nondegenerate since $N''(z) = -2e^z \sin z \neq 0$ there).
- Singularities: none (entire).

Real/imag parts and modulus. For $z = x + iy$,

$$e^z = e^x(\cos y + i \sin y), \qquad \cos(x+iy) = \cos x \cosh y - i \sin x \sinh y,$$

so

$$\begin{aligned} N(z) = e^x \Big(&\cos y \cos x \cosh y + \sin y \sin x \sinh y\Big) \\ &+ i\, e^x \Big(\sin y \cos x \cosh y - \cos y \sin x \sinh y\Big), \end{aligned}$$

and

$$|N(x+iy)|^2 = e^{2x}\Big(\cos^2 x\, \cosh^2 y + \sin^2 x\, \sinh^2 y\Big).$$

Vertical lines (x fixed). As $|y| \to \infty$, $\cosh y \sim \sinh y \sim \frac{1}{2}e^{|y|}$, hence

$$|N(x+iy)| \sim \tfrac{1}{2}\, e^{x+|y|} \qquad (|y| \to \infty).$$

Horizontal lines (y fixed). Using $|\cos(x+iy)| \in \big[|\sinh y|, |\cosh y|\big]$ as x varies,

$$|N(x+iy)| \in \big[e^x|\sinh y|,\ e^x|\cosh y|\big],$$

so $|N(x+iy)|$ oscillates in x with an e^x envelope (growth for $x \to +\infty$, decay for $x \to -\infty$).

Directional growth (indicator). Along $z = re^{i\theta}$,

$$\log|N(re^{i\theta})| \approx r\ \max\{\cos\theta + \sin\theta,\ \cos\theta - \sin\theta\},$$

so the indicator is $h(\theta) = \max\{\cos\theta + \sin\theta,\ \cos\theta - \sin\theta\}$, with maximum $\sqrt{2}$ at $\theta = \pm\frac{\pi}{4}$.

Series at 0.

$$e^z \cos z = 1 + z + \underbrace{0}_{z^2} + \frac{z^3}{3} + O(z^4),$$

implying, for example,

$$\int_{|z|=1} \frac{e^z \cos z}{z^3}\, dz = 2\pi i\, [z^{-1}]\Big(\frac{e^z \cos z}{z^3}\Big) = 0.$$

Exponential integrands on the unit circle

Anatomy of $E(z) = e^z$

- **Entire/type.** E is entire of order 1 and exponential type 1.
- **Zeros/critical points.** No zeros; $E'(z) = e^z \neq 0$ everywhere, so no critical points.
- **Modulus and lines.** For $z = x + iy$, $|E(z)| = e^x$. Along vertical lines (x fixed) the modulus is constant and the argument increases linearly with y. Along horizontal lines (y fixed) the modulus is e^x (exponential growth/decay).
- **Directional growth.** On the ray $z = re^{i\theta}$, $|E(re^{i\theta})| = e^{r\cos\theta}$; indicator $h(\theta) = \cos\theta$.
- **Series at 0.** $e^z = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \cdots$.

Cauchy evaluations on $\gamma = \{|z| = 1\}$

Let $f(z) = e^z$.

(a) Using the Cauchy Integral Formula with $a = 0$,

$$\int_\gamma \frac{e^z}{z}\, dz = 2\pi i\, f(0) = 2\pi i.$$

(b) Using the higher-order Cauchy formula with $n = 2$,

$$\int_\gamma \frac{e^z}{z^3}\, dz = 2\pi i\, \frac{f^{(2)}(0)}{2!} = 2\pi i \cdot \frac{1}{2} = \pi i.$$

(Here $f^{(2)}(0) = e^0 = 1$. Equivalently, the z^{-1}-coefficient in the Laurent expansion of e^z/z^3 is $1/2$.)

Problem

Let γ be the positively oriented unit circle $\{|z| = 1\}$. Use the Cauchy Integral Formula to evaluate

$$\text{(a)} \quad \int_\gamma \frac{e^z \cos z}{z}\, dz, \qquad \text{(b)} \quad \int_\gamma \frac{e^z \cos z}{z^3}\, dz.$$

Background and method

We recall two standard versions of the Cauchy Integral Formula.

Cauchy Integral Formula (CIF). If f is holomorphic in a neighborhood of the closed contour γ and a is a point inside γ, then

$$f(a) = \frac{1}{2\pi i} \oint_\gamma \frac{f(\zeta)}{\zeta - a}\, d\zeta.$$

Equivalently,

$$\oint_\gamma \frac{f(\zeta)}{\zeta - a}\, d\zeta = 2\pi i\, f(a).$$

Higher-order Cauchy formula. If f is holomorphic near a, then for $n \geq 0$,

$$f^{(n)}(a) = \frac{n!}{2\pi i} \oint_\gamma \frac{f(\zeta)}{(\zeta - a)^{n+1}}\, d\zeta,$$

or equivalently,

$$\oint_\gamma \frac{f(\zeta)}{(\zeta - a)^{n+1}}\, d\zeta = 2\pi i\, \frac{f^{(n)}(a)}{n!}.$$

—

Structure of the integrand

Let

$$f(z) = e^z \cos z.$$

Then f is *entire*, since both e^z and $\cos z$ are entire.

Zeros. $e^z \neq 0$ everywhere, and $\cos z = 0$ at $z = \frac{\pi}{2} + k\pi$, $k \in \mathbb{Z}$. None of these zeros lie on $|z| = 1$, so f has no singularities on γ.

Singularity structure of the integrands.

$$\frac{f(z)}{z} \text{ has a simple pole at } z = 0, \qquad \frac{f(z)}{z^3} \text{ has a triple pole at } z = 0.$$

Since $f(0) = e^0 \cos 0 = 1 \neq 0$, the pole orders are not reduced.

—

(a) Evaluation of $\displaystyle\int_\gamma \frac{e^z \cos z}{z}\, dz$

By the Cauchy Integral Formula with $a = 0$, we have

$$\oint_\gamma \frac{f(z)}{z}\, dz = 2\pi i\, f(0).$$

Compute $f(0)$:

$$f(0) = e^0 \cos 0 = 1.$$

Therefore,

$$\boxed{\int_\gamma \frac{e^z \cos z}{z}\, dz = 2\pi i.}$$

—

(b) Evaluation of $\displaystyle\int_\gamma \frac{e^z \cos z}{z^3}\, dz$

By the higher-order Cauchy formula with $a = 0$ and $n = 2$,

$$\oint_\gamma \frac{f(z)}{z^3}\, dz = 2\pi i\, \frac{f^{(2)}(0)}{2!}.$$

Compute the derivatives.

$$f'(z) = e^z(\cos z - \sin z), \qquad f''(z) = e^z(\cos z - \sin z) + e^z(-\sin z - \cos z) = -2e^z \sin z.$$

Thus,

$$f''(0) = -2e^0 \sin 0 = 0.$$

Hence

$$\boxed{\int_\gamma \frac{e^z \cos z}{z^3}\, dz = 0.}$$

—

Alternative check using the series expansion

We can verify the result directly from the Maclaurin series.

$$e^z = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \cdots, \qquad \cos z = 1 - \frac{z^2}{2} + \frac{z^4}{24} - \cdots.$$

Multiplying,

$$e^z \cos z = (1 + z + \tfrac{z^2}{2} + \tfrac{z^3}{6} + \cdots)(1 - \tfrac{z^2}{2} + \tfrac{z^4}{24} - \cdots) = 1 + z + 0 \cdot z^2 + \tfrac{1}{3}z^3 + O(z^4).$$

Then

$$\frac{e^z \cos z}{z^3} = \frac{1}{z^3} + \frac{1}{z^2} + 0 \cdot \frac{1}{z} + \tfrac{1}{3} + O(z),$$

and the coefficient of $\frac{1}{z}$ is zero. By Cauchy's theorem (or the residue theorem),

$$\oint_\gamma \frac{e^z \cos z}{z^3}\, dz = 2\pi i \, [z^{-1}] \left(\frac{e^z \cos z}{z^3} \right) = 0.$$

—

Summary and interpretation

$$\boxed{\begin{array}{ll} (a) & \displaystyle\int_\gamma \frac{e^z \cos z}{z}\, dz = 2\pi i, \\ (b) & \displaystyle\int_\gamma \frac{e^z \cos z}{z^3}\, dz = 0. \end{array}}$$

Geometric intuition. - The integrands have their only singularity at the origin, entirely enclosed by the contour γ. - The first integral measures the value of $f(0)$ via the Cauchy kernel $1/z$. - The second integral involves the second derivative of f at 0; since $f''(0) = 0$, its contribution vanishes. - In the Laurent expansion viewpoint, the integral picks out the coefficient of $\frac{1}{z}$ in $\frac{f(z)}{z^3}$, which here is zero.

Mean value property and maximum modulus principle

Problem (exact)

Let $f : U \to \mathbb{C}$ be holomorphic on a domain U.

(a) Show that, if $\overline{B}(a, r) \subset U$, then

$$f(a) = \frac{1}{2\pi} \int_0^{2\pi} f(a + re^{it})\, dt.$$

(b) Suppose $a \in U$ is such that $|f(a)| = \max_U |f|$. Show that $|f|$ must be constant near a.

(c) Deduce that f is constant on all of U. *Hint:* Consider $S = \{z \in U : |f(z)| = |f(a)|\}$.

Background

Cauchy integral formula on circles. If $\overline{B}(a,r) \subset U$, then

$$f(a) = \frac{1}{2\pi i} \int_{|\zeta - a| = r} \frac{f(\zeta)}{\zeta - a}\, d\zeta.$$

With $\zeta = a + re^{it}$ and $d\zeta = ire^{it}dt$, this becomes the circular average.

Equality in the triangle inequality. For an integrable complex function g, $\left|\int g\right| \le \int |g|$ with equality iff g has a constant argument almost everywhere on the path of integration.

Cauchy integral for derivatives. If f is holomorphic on $\overline{B}(a,r)$, then for $n \ge 1$,

$$f^{(n)}(a) = \frac{n!}{2\pi i} \int_{|\zeta - a| = r} \frac{f(\zeta)}{(\zeta - a)^{n+1}}\, d\zeta.$$

Solution

(a) Mean value property. By Cauchy's formula,

$$f(a) = \frac{1}{2\pi i} \int_{|\zeta - a| = r} \frac{f(\zeta)}{\zeta - a}\, d\zeta = \frac{1}{2\pi i} \int_0^{2\pi} \frac{f(a + re^{it})}{re^{it}}\, (ire^{it})\, dt = \frac{1}{2\pi} \int_0^{2\pi} f(a + re^{it})\, dt.$$

(b) Local constancy at an interior maximum. Assume $|f(a)| = \max_U |f|$. For $r > 0$ with $\overline{B}(a,r) \subset U$,

$$|f(a)| = \left| \frac{1}{2\pi} \int_0^{2\pi} f(a + re^{it})\, dt \right| \le \frac{1}{2\pi} \int_0^{2\pi} |f(a + re^{it})|\, dt \le \frac{1}{2\pi} \int_0^{2\pi} |f(a)|\, dt = |f(a)|.$$

Thus equality holds throughout. Hence $|f(a + re^{it})| = |f(a)|$ and $f(a + re^{it})$ has a constant argument (the same as $f(a)$) for almost every t. Therefore

$$f(a + re^{it}) = f(a) \quad \text{for all } t,$$

i.e. f is constant on the circle $|z - a| = r$. For such an r, Cauchy's formula for $n \ge 1$ gives

$$f^{(n)}(a) = \frac{n!}{2\pi i} \int_{|\zeta - a| = r} \frac{f(\zeta)}{(\zeta - a)^{n+1}}\, d\zeta = \frac{n!}{2\pi i} f(a) \int_{|\zeta - a| = r} \frac{1}{(\zeta - a)^{n+1}}\, d\zeta = 0.$$

Hence the Taylor series of f at a is the constant $f(a)$; there exists $\rho > 0$ such that $f \equiv f(a)$ on $B(a,\rho)$. In particular, $|f|$ is constant near a.

(c) Constancy on the whole domain. Let $M = |f(a)|$ and $S = \{z \in U : |f(z)| = M\}$. The set S is closed (preimage of $\{M\}$ under the continuous map $z \mapsto |f(z)|$). By part (b), for each $z \in S$ there exists a neighborhood on which f is constant, hence that neighborhood lies in S; thus S is open. Since U is connected and $S \neq \varnothing$, we conclude $S = U$, so $|f|$ is constant on U, and by (b) (applied at any point) f is constant on U.

Examples

- For $f(z) = z^n$, $f(0) = 0 = \dfrac{1}{2\pi}\displaystyle\int_0^{2\pi} (re^{it})^n dt$ (mean value).
- For $f(z) = e^z$ on any disk, $\max_{\overline{B}(a,r)} |f|$ occurs on $|z - a| = r$ (no interior maximum unless f is constant).

Mean Value Property for Holomorphic and Harmonic Functions

Statement (Holomorphic Case)

Let f be holomorphic on a domain $U \subset \mathbb{C}$. If the closed disk $\overline{B}(a, r) \subset U$, then

$$\boxed{f(a) \; = \; \frac{1}{2\pi} \int_0^{2\pi} f\big(a + re^{it}\big)\, dt}$$

i.e. the value of f at the center equals the average of its values on the surrounding circle.

One-Line Proof

By Cauchy's integral formula on the circle $|\zeta - a| = r$,

$$f(a) = \frac{1}{2\pi i} \int_{|\zeta - a| = r} \frac{f(\zeta)}{\zeta - a}\, d\zeta.$$

Parametrize $\zeta = a + re^{it}$ so that $d\zeta = ire^{it}\, dt$. Then

$$f(a) = \frac{1}{2\pi i} \int_0^{2\pi} \frac{f(a + re^{it})}{re^{it}} (ire^{it})\, dt = \frac{1}{2\pi} \int_0^{2\pi} f(a + re^{it})\, dt.$$

Harmonic Version

If u is harmonic on U and $\overline{B}(a,r) \subset U$, then

$$\boxed{u(a) \;=\; \frac{1}{2\pi}\int_0^{2\pi} u\big(a+re^{it}\big)\,dt}$$

(For $f = u + iv$ holomorphic, both u and v are harmonic and satisfy this property.)

Intuition and Consequences

Rigidity / No interior spikes. The center value is a circular average; thus a modulus maximum cannot occur inside unless f is constant (maximum modulus principle).

Cauchy estimates. From circular averages and Cauchy formulas for derivatives, one gets the standard bounds

$$|f^{(n)}(a)| \le \frac{n!}{r^n} \max_{|\zeta-a|=r} |f(\zeta)|.$$

Uniqueness. For harmonic functions, boundary data determines the interior via Poisson's formula; the mean value property is a key equivalent.

Examples

1) Polynomials. For $f(z) = z^n$ and $a = 0$,

$$\frac{1}{2\pi}\int_0^{2\pi} (re^{it})^n\,dt = \begin{cases} 1, & n = 0, \\ 0, & n \ge 1, \end{cases} \;=\; f(0).$$

2) Exponential. Let $f(z) = e^z$. Then on $|z-a| = r$,

$$f(a+re^{it}) = e^a\,e^{re^{it}} = e^a \sum_{k=0}^{\infty} \frac{r^k e^{ikt}}{k!}.$$

Averaging over $t \in [0, 2\pi]$ kills all $k \ge 1$, leaving

$$\frac{1}{2\pi}\int_0^{2\pi} f(a+re^{it})\,dt = e^a = f(a).$$

3) Harmonic example. If $u(x,y)=x$ (harmonic), then for any circle centered at $a=x_0+iy_0$,

$$\frac{1}{2\pi}\int_0^{2\pi} u\big(a+re^{it}\big)\,dt=\frac{1}{2\pi}\int_0^{2\pi}(x_0+r\cos t)\,dt=x_0=u(a).$$

Equality Case and Maximum Modulus

If $|f(a)|=\max_{\overline{B}(a,r)}|f|$, then

$$|f(a)|=\left|\frac{1}{2\pi}\int_0^{2\pi} f(a+re^{it})\,dt\right|\le\frac{1}{2\pi}\int_0^{2\pi}|f(a+re^{it})|\,dt\le|f(a)|.$$

Equality forces $f(a+re^{it})$ to have *constant argument* and constant modulus on the circle, hence f is constant on that circle. Cauchy's formula for derivatives then yields $f^{(n)}(a)=0$ for $n\ge 1$, so f is constant near a; on a connected domain, f is constant everywhere.

Harmonic vs. Holomorphic: What Implies What?

Holomorphic $\Rightarrow$ Harmonic

Let $f=u+iv$ be holomorphic on $U\subset\mathbb{C}$. Then $u,v\in C^\infty(U)$ and satisfy the Cauchy–Riemann equations

$$u_x=v_y,\qquad u_y=-\,v_x.$$

Differentiating and adding,

$$u_{xx}+u_{yy}=0,\qquad v_{xx}+v_{yy}=0,$$

so u and v are *harmonic*. Hence every holomorphic function has harmonic real and imaginary parts.

Harmonic $\not\Rightarrow$ Holomorphic (in general)

A real-valued harmonic function u is not itself holomorphic (nonconstant real-valued holomorphic maps do not exist by the Open Mapping Theorem).

For a complex-valued $f=u+iv$ to be holomorphic one needs both:

$$\text{(i) } \Delta u=\Delta v=0\quad\text{and}\quad\text{(ii) } u_x=v_y,\ u_y=-v_x.$$

Harmonicity alone does not enforce the Cauchy–Riemann relations.

Harmonic Conjugates and Topology

Given harmonic u on U, define the 1-form

$$\alpha := -\,u_y\,dx + u_x\,dy.$$

Then $d\alpha = (u_{xx} + u_{yy})\,dx \wedge dy = \Delta u\,dx \wedge dy = 0$, so α is *closed.*

Local existence. Closed $\Rightarrow$ exact locally: there exists v (locally) with $dv = \alpha$, i.e.

$$v_y = u_x, \qquad v_x = -u_y.$$

Thus $f = u + iv$ is *locally* holomorphic.

Global existence. If U is *simply connected*, closed $\Rightarrow$ exact globally, so there is a global harmonic conjugate v and $f = u + iv$ is globally holomorphic (unique up to an additive constant in v). If U is not simply connected, a global v exists iff all periods of α vanish:

$$\oint_\gamma \alpha \;=\; 0 \quad \text{for every closed loop } \gamma \subset U.$$

Examples

Holomorphic $\Rightarrow$ harmonic. $f(z) = e^z$; then $u = e^x \cos y$ and $v = e^x \sin y$ satisfy $\Delta u = \Delta v = 0$.

Harmonic but no global holomorphic primitive on $\mathbb{C} \setminus \{0\}$. $u(z) = \log|z|$ is harmonic on $\mathbb{C} \setminus \{0\}$. Locally $u = \Re(\log z)$, but a global conjugate would be $\arg z$, which is multivalued; the periods are nonzero, so no global single-valued holomorphic F with $\Re F = u$.

Harmonic with a global conjugate. $u(x, y) = x^2 - y^2$ is harmonic; $v(x, y) = 2xy$ satisfies the CR relations, and $f = u + iv = (x + iy)^2 = z^2$ is holomorphic.

Harmonic components but not holomorphic. $f(z) = \overline{z} = x - iy$. Here $u = x$, $v = -y$ are harmonic, but $u_x = 1 \neq v_y = -1$; the CR equations fail, so f is not holomorphic (it is anti-holomorphic).

Bottom Line

- **Always:** holomorphic $\Rightarrow$ (real and imaginary parts) harmonic.
- **Locally:** harmonic u is the real part of a holomorphic function.
- **Globally:** harmonic u is the real part of a holomorphic function *iff* the domain is simply connected (or, equivalently, the periods of $-u_y\,dx + u_x\,dy$ vanish).
- **Not sufficient:** harmonicity alone does not imply holomorphicity; the Cauchy–Riemann equations are the extra condition.

Product-of-distances lower bound on the unit circle

Problem

Let $w_1, \dots, w_n$ lie on the unit circle $\{|z| = 1\}$. Show that there exists z on the unit circle such that

$$\prod_{k=1}^{n} |z - w_k| \;\geq\; 1.$$

Key observation: average of $\log|z - w|$ is zero on the circle

Fix w with $|w| = 1$ and parametrize the circle by $z = e^{it}$, $t \in [0, 2\pi]$. Writing $w = e^{i\phi}$,

$$|e^{it} - w| = \left|e^{it} - e^{i\phi}\right| = 2\left|\sin\frac{t-\phi}{2}\right|.$$

Hence

$$\frac{1}{2\pi}\int_0^{2\pi} \log|e^{it} - w|\,dt = \frac{1}{2\pi}\int_0^{2\pi} \log\!\Big(2\Big|\sin\frac{t-\phi}{2}\Big|\Big)\,dt = \frac{1}{2\pi}\int_0^{2\pi} \log\!\Big(2\Big|\sin\frac{u}{2}\Big|\Big)\,du = 0,$$

where the shift $u = t - \phi$ and the classical integral $\displaystyle\int_0^{2\pi} \log\!\Big(2\sin\frac{u}{2}\Big)\,du = 0$ were used.

Proof of the claim

Define the polynomial

$$P(z) = \prod_{k=1}^{n} (z - w_k).$$

For $z = e^{it}$ on the unit circle,

$$\log |P(e^{it})| = \sum_{k=1}^{n} \log |e^{it} - w_k|.$$

Averaging over $t \in [0, 2\pi]$ and using the previous calculation term-by-term,

$$\frac{1}{2\pi} \int_0^{2\pi} \log |P(e^{it})| \, dt = \sum_{k=1}^{n} \frac{1}{2\pi} \int_0^{2\pi} \log |e^{it} - w_k| \, dt = 0.$$

If $\log |P(e^{it})| < 0$ for all t, the average would be negative. Therefore there exists t_0 with

$$\log |P(e^{it_0})| \;\geq\; 0 \quad \iff \quad |P(e^{it_0})| = \prod_{k=1}^{n} |e^{it_0} - w_k| \;\geq\; 1.$$

Setting $z = e^{it_0}$ completes the proof.

Remarks and variants

- Equivalently, by Jensen's formula applied to P (whose zeros lie on $|z| = 1$), the mean of $\log |P(e^{it})|$ over the unit circle equals $\log |P(0)| = 0$, since $|P(0)| = \prod |w_k| = 1$.
- For $n = 1$, choosing $z = -w_1$ gives $|z - w_1| = 2 \geq 1$. For general n, the statement guarantees at least one z with product ≥ 1, though the maximizing z depends on the configuration of the w_k.

Entire function with nonnegative real part is constant

Problem

Let f be holomorphic on $\mathbb{C}$ and suppose $\operatorname{Re} f(z) \geq 0$ for all $z \in \mathbb{C}$. Show that f is constant. *(Hint: consider* $\exp(-f(z))$*.)*

Background

- **Liouville's Theorem.** Every bounded entire function is constant.
- If $\operatorname{Re} f \geq 0$ and $g(z) := e^{-f(z)}$, then

  $$|g(z)| = |e^{-f(z)}| = e^{-\operatorname{Re} f(z)} \leq 1,$$

  so g is entire and bounded.

Solution

Define $g(z) := e^{-f(z)}$. Then g is entire and satisfies $|g(z)| \le 1$ for all $z \in \mathbb{C}$. By Liouville's theorem, g is constant: $g(z) \equiv c$ with $|c| \le 1$.

Hence $e^{-f(z)} \equiv c$. Choosing any logarithm of the constant c,

$$-f(z) \equiv \log c \qquad \Longrightarrow \qquad f(z) \equiv -\log c,$$

so f is constant on $\mathbb{C}$.

Remark (optional)

If f were nonconstant, then $g = e^{-f}$ would be nonconstant and thus an open map. But $g(\mathbb{C}) \subset \{w : |w| \le 1\}$ is not open, a contradiction. Therefore f must be constant.

Reverse assumption: $\operatorname{Re} f \le 0$ implies f is constant

Claim

If f is entire and $\operatorname{Re} f(z) \le 0$ for all $z \in \mathbb{C}$, then f is constant.

Proof

Define $h(z) := e^{f(z)}$. Then h is entire and

$$|h(z)| = |e^{f(z)}| = e^{\operatorname{Re} f(z)} \le 1 \qquad (\forall z \in \mathbb{C}).$$

Hence h is a bounded entire function. By Liouville's theorem, h is constant: $h(z) \equiv c$ with $|c| \le 1$. Taking any logarithm of the constant c, we get $f(z) \equiv \log c$, a constant function.

Remark (generalization)

If $\operatorname{Re} f$ is bounded *above* or *below* on $\mathbb{C}$, then f is constant. Indeed, use e^f when $\operatorname{Re} f \le M$ or e^{-f} when $\operatorname{Re} f \ge m$ to obtain a bounded entire function and apply Liouville.

Half–plane criteria for constancy of entire functions

General Lemma (rotation–shift half–plane test)

Let f be entire. Suppose there exist $\theta \in \mathbb{R}$ and $c \in \mathbb{R}$ such that

$$\Re\big(e^{-i\theta} f(z)\big) \;\geq\; c \qquad (\forall z \in \mathbb{C}).$$

Then f is constant.

Proof. Define

$$g(z) := \exp\big(-\, e^{-i\theta}\,(f(z) - c)\big).$$

Then

$$|g(z)| = \exp\Big(-\,\Re\big(e^{-i\theta}(f(z) - c)\big)\Big) \;\leq\; 1,$$

so g is bounded and entire. By Liouville's theorem, g is constant; hence f is constant. □

Immediate corollaries (choose θ appropriately)

- **Real part bounded below.** If $\Re f(z) \geq a$ for all z, take $\theta = 0$:

$$|e^{-(f-a)}| = e^{-(\Re f - a)} \leq 1 \;\Rightarrow\; e^{-(f-a)} \text{ constant } \;\Rightarrow\; f \text{ constant.}$$

- **Real part bounded above.** If $\Re f(z) \leq a$ for all z, use e^{f-a}:

$$|e^{f-a}| = e^{\Re f - a} \leq 1 \;\Rightarrow\; e^{f-a} \text{ constant } \;\Rightarrow\; f \text{ constant.}$$

- **Imaginary part bounded below.** If $\Im f(z) \geq b$ for all z, take $\theta = \frac{\pi}{2}$:

$$|e^{if}| = e^{-\Im f} \leq e^{-b} \;\Rightarrow\; e^{if} \text{ constant } \;\Rightarrow\; f \text{ constant.}$$

- **Imaginary part bounded above.** If $\Im f(z) \leq b$ for all z, take $\theta = -\frac{\pi}{2}$:

$$|e^{-if}| = e^{\Im f} \leq e^{b} \;\Rightarrow\; e^{-if} \text{ constant } \;\Rightarrow\; f \text{ constant.}$$

- **Tilted half–planes.** If $\Re(e^{-i\theta} f(z)) \geq a$ (or $\leq a$) for some fixed θ, apply the lemma directly. Thus any entire map with image contained in a rotated/translated half–plane is constant.

Strip versions and harmonic viewpoint

If $a \le \Re(e^{-i\theta} f) \le b$ on $\mathbb{C}$, then $\Re(e^{-i\theta} f)$ is a bounded harmonic function on $\mathbb{C}$, hence constant. The Cauchy–Riemann equations then force the imaginary part to be constant as well, so f is constant.

Similarly, if $a \le \Im(e^{-i\theta} f) \le b$, apply the same argument to the harmonic function $\Im(e^{-i\theta} f)$.

Remarks

- The choices $e^{\pm f}$ and $e^{\pm if}$ are tailored so that their moduli are $e^{\pm \Re f}$ and $e^{\mp \Im f}$, respectively.
- This gives a unified proof that an entire function with image in *any* (open or closed) half–plane must be constant.
- Mapping a half–plane to the unit disk with a Möbius map (e.g. Cayley transform) yields the same conclusion: the composition is bounded entire, hence constant.

Growth and domination for entire functions

Problem

Let f be an entire function.

(i) Prove that f is a polynomial of degree at most k if and only if there exist real constants $M, R > 0$ and an integer k such that

$$|f(z)| \le M\,|z|^k \qquad \text{for } |z| > R.$$

(ii) Let g be an entire function such that $|f(z)| \le |g(z)|$ for all $z \in \mathbb{C}$. Show that there exists $c \in \mathbb{C}$ such that $f(z) = c\,g(z)$ for all $z \in \mathbb{C}$.

Background

Cauchy estimates. If f is holomorphic on $|z| \le \rho$, then

$$f^{(n)}(0) = \frac{n!}{2\pi i} \int_{|z|=\rho} \frac{f(z)}{z^{n+1}}\,dz, \qquad |f^{(n)}(0)| \le \frac{n!}{\rho^n} \max_{|z|=\rho} |f(z)|.$$

Liouville and removable singularities. A bounded entire function is constant. If h is holomorphic on a punctured neighborhood of a and bounded near a, then h extends holomorphically to a.

Solution to (i)

"Only if." If $f(z) = a_k z^k + \cdots + a_0$ has degree $\le k$, then for $|z| > 1$,

$$|f(z)| \le \big(|a_k| + \cdots + |a_0|\big)\, |z|^k.$$

Taking $R \ge 1$ and $M = \sum_{j=0}^{k} |a_j|$ gives the claim.

"If." Assume there exist $M, R > 0$ and integer k with $|f(z)| \le M|z|^k$ for $|z| > R$. Fix $\rho > R$. By Cauchy's estimate,

$$|f^{(n)}(0)| \le \frac{n!}{\rho^n} \max_{|z|=\rho} |f(z)| \le \frac{n!}{\rho^n}\, M\, \rho^k = M\, n!\, \rho^{k-n}.$$

If $n > k$, then letting $\rho \to \infty$ yields $f^{(n)}(0) = 0$. Hence all derivatives of order $> k$ vanish, so f is a polynomial of degree at most k.

Solution to (ii)

If $g \equiv 0$, then $|f| \le |g|$ forces $f \equiv 0$, and $f = cg$ is trivial. Assume $g \not\equiv 0$.

Define $h(z) = \dfrac{f(z)}{g(z)}$ where $g(z) \ne 0$. We show h extends to an entire, bounded function.

Let a be a zero of g of order $m \ge 1$, so $g(z) = (z-a)^m g_m(z)$ with g_m holomorphic and $g_m(a) \ne 0$. From $|f| \le |g|$ we have, for z near a,

$$|f(z)| \le C\, |z-a|^m$$

for some $C > 0$, hence $f(z) = (z-a)^m f_m(z)$ with f_m holomorphic (and bounded) near a. Then on the punctured neighborhood,

$$h(z) = \frac{f(z)}{g(z)} = \frac{f_m(z)}{g_m(z)}.$$

Since $g_m(a) \ne 0$ and f_m is bounded near a, h is bounded near a; by the removable singularities theorem, h extends holomorphically to a. Doing this at all zeros of g, we obtain an entire extension of h. Moreover, on the set where $g \ne 0$ we have $|h(z)| = \dfrac{|f(z)|}{|g(z)|} \le 1$, therefore by continuity $|h| \le 1$ everywhere.

Thus h is bounded and entire, so by Liouville, $h \equiv c$ for some $c \in \mathbb{C}$ (with $|c| \le 1$). Consequently $f(z) = c\, g(z)$ for all $z \in \mathbb{C}$.

Examples

- If $f(z) = 3z^5 - 2z + 7$ then $|f(z)| \le (3+2+7)\,|z|^5$ for $|z| > 1$; the growth condition holds with $k = 5$.
- If $|f(z)| \le |\sin z|$ on $\mathbb{C}$, then by (ii) there is $c \in \mathbb{C}$ with $f(z) = c\sin z$ for all z.

Poisson and Schwarz Integral Formulas on the Unit Disk

Let $f = u + iv$ be holomorphic in a domain containing the closed unit disk $\overline{\mathbb{D}}$.

(a) Show that

$$u(0) = \frac{1}{2\pi}\int_0^{2\pi} u(e^{it})\,dt.$$

(b) Assume that for every $z_0 \in \mathbb{D}$ the map

$$h_{z_0}(z) = \frac{z + z_0}{1 + \overline{z_0}\,z}$$

is a bijection of $\mathbb{D}$ onto itself. Using this map, show that for $z \in \mathbb{D}$,

$$u(z) = \frac{1}{2\pi}\int_0^{2\pi} u(e^{it})\,\frac{1 - |z|^2}{|e^{it} - z|^2}\,dt.$$

(c) Using the fact that

$$\frac{|w|^2 - |z|^2}{|z - w|^2} = \Re\left(\frac{w + z}{w - z}\right),$$

show that

$$f(z) = \frac{1}{2\pi}\int_0^{2\pi} u(e^{it})\,\frac{e^{it} + z}{e^{it} - z}\,dt \;+\; i\,v(0).$$

Background (first theory)

Mean value property. If u is harmonic on a neighborhood of $\overline{\mathbb{D}}$, then for $0 < r \le 1$,

$$u(0) = \frac{1}{2\pi}\int_0^{2\pi} u(re^{it})\,dt.$$

Poisson kernel on $\mathbb{D}$. For $z = re^{i\phi} \in \mathbb{D}$ and boundary point e^{it},

$$P_r(\phi - t) = \frac{1 - r^2}{|e^{it} - re^{i\phi}|^2} = \frac{1 - |z|^2}{|e^{it} - z|^2}.$$

If $g \in C(\partial\mathbb{D})$ and U is harmonic in $\mathbb{D}$ with nontangential boundary values g, then

$$U(z) = \frac{1}{2\pi}\int_0^{2\pi} g(e^{it})\,P_r(\phi - t)\,dt.$$

Disk automorphisms. For $z_0 \in \mathbb{D}$,

$$h_{z_0}(z) = \frac{z + z_0}{1 + \overline{z_0} z}, \qquad h_{z_0}(0) = z_0, \qquad h_{z_0}^{-1} = h_{-z_0},$$

is a Möbius automorphism of $\mathbb{D}$ carrying $\partial\mathbb{D}$ onto itself and preserving harmonic measure.

Schwarz/Herglotz kernel. For $z \in \mathbb{D}$,

$$K(z, e^{it}) = \frac{e^{it} + z}{e^{it} - z}, \qquad \Re K(z, e^{it}) = \frac{1 - |z|^2}{|e^{it} - z|^2} = P_r(\phi - t).$$

Solutions

(a) Mean value at the center

Since f is holomorphic near $\overline{\mathbb{D}}$, its real part u is harmonic there. The mean value property at the origin gives

$$u(0) = \frac{1}{2\pi} \int_0^{2\pi} u(e^{it})\, dt.$$

(b) Poisson integral via h_{z_0}

Fix $z \in \mathbb{D}$ and put $z_0 = z$. Consider the harmonic function

$$U(\zeta) := u\big(h_z(\zeta)\big), \qquad \zeta \in \mathbb{D}.$$

By (a), applied at the origin for U,

$$U(0) = \frac{1}{2\pi} \int_0^{2\pi} U(e^{it})\, dt = \frac{1}{2\pi} \int_0^{2\pi} u\big(h_z(e^{it})\big)\, dt.$$

But $U(0) = u(h_z(0)) = u(z)$. Using the boundary change of variables $e^{i\theta} = h_z(e^{it})$ and the known identity

$$\frac{dt}{d\theta} = \frac{1 - |z|^2}{|e^{i\theta} - z|^2},$$

we obtain

$$u(z) = \frac{1}{2\pi} \int_0^{2\pi} u(e^{i\theta})\, \frac{1 - |z|^2}{|e^{i\theta} - z|^2}\, d\theta,$$

which is the Poisson integral formula.

(c) Schwarz integral for f

Define

$$F(z) := \frac{1}{2\pi} \int_0^{2\pi} u(e^{it})\, \frac{e^{it} + z}{e^{it} - z}\, dt.$$

For fixed $z \in \mathbb{D}$ the integrand is holomorphic in z and integrable in t, so F is holomorphic on $\mathbb{D}$. Taking real parts and using

$$\Re\left(\frac{e^{it}+z}{e^{it}-z}\right) = \frac{1-|z|^2}{|e^{it}-z|^2},$$

we get

$$\Re F(z) = \frac{1}{2\pi}\int_0^{2\pi} u(e^{it})\,\frac{1-|z|^2}{|e^{it}-z|^2}\,dt = u(z)$$

by part (b). Hence $F - f$ is holomorphic with $\Re(F-f) \equiv 0$ on $\mathbb{D}$, so $F - f \equiv iC$ for some real constant C. Evaluating at $z = 0$,

$$F(0) = \frac{1}{2\pi}\int_0^{2\pi} u(e^{it})\,dt = u(0),$$

so

$$u(0) + iC = f(0) = u(0) + iv(0) \quad \Rightarrow \quad C = v(0).$$

Therefore

$$f(z) = F(z) + i\,v(0) = \frac{1}{2\pi}\int_0^{2\pi} u(e^{it})\,\frac{e^{it}+z}{e^{it}-z}\,dt + i\,v(0).$$

Normalization and quick intuition

Normalize once:

$$K_t(z) = C\big(e^{-it}z\big), \qquad C(\zeta) = \frac{1+\zeta}{1-\zeta} \quad \text{(Cayley transform)}.$$

So every geometric fact for K_t is a rotation of the corresponding fact for C (which "focuses" at the boundary point 1).

Natural domain and image

$$\boxed{\mathbb{D} = \{\, z : |z| < 1 \,\}}, \qquad \boxed{K_t(\mathbb{D}) = \{\, w : \operatorname{Re} w > 0 \,\} \text{ (right half-plane)}}.$$

Reason: for $|z| < 1$,

$$\operatorname{Re} K_t(z) = \frac{1-|z|^2}{|e^{it}-z|^2} > 0.$$

Boundary behavior.

$$K_t(0) = 1, \qquad \lim_{z \to e^{it}} K_t(z) = \infty, \qquad K_t\big(\partial\mathbb{D} \setminus \{e^{it}\}\big) = i\mathbb{R}.$$

Thus $K_t : \mathbb{D} \to \{\operatorname{Re} w > 0\}$ is a conformal bijection, sending the boundary point e^{it} to ∞ and preserving orientation.

Inverse and derivative (for checks)

$$K_t^{-1}(w) = e^{it}\,\frac{w-1}{w+1}, \qquad K_t'(z) = \frac{2e^{it}}{(e^{it}-z)^2}, \qquad \left|K_t'(z)\right| = \frac{2}{|e^{it}-z|^2}.$$

Vertical lines in the image and their preimages

Fix $c > 0$. The vertical line

$$L_c = \{\, w : \ \operatorname{Re} w = c \,\}$$

pulls back to the level set

$$\operatorname{Re} K_t(z) = \frac{1-|z|^2}{|e^{it}-z|^2} = c.$$

Rotate by $u = e^{-it}z$ (so $e^{it} \mapsto 1$). For the Cayley map C this is

$$\operatorname{Re} C(u) = \frac{1-|u|^2}{|1-u|^2} = c \iff |1-u|^2 = c^{-1}(1-|u|^2).$$

Writing $u = x+iy$ gives the circle equation

$$\boxed{(x - \tfrac{1}{1+c})^2 + y^2 = \left(\tfrac{1}{1+c}\right)^2}.$$

Hence in the u–plane we get a circle of radius $1/(1+c)$ centered at $1/(1+c)$ on the real axis, tangent to $\partial\mathbb{D}$ at $u = 1$. Rotating back:

$$\boxed{\{\operatorname{Re} K_t(z) = c\} \text{ is a circle in } \mathbb{D} \text{ tangent to } \partial\mathbb{D} \text{ at } \tau = e^{it}.}$$

As $c \downarrow 0$ the circle expands to $\partial\mathbb{D}$ (except at τ); as $c \uparrow \infty$ it shrinks to τ.

Vertical bands. The strip $a < \operatorname{Re} w < b$ corresponds to the lens-like region in $\mathbb{D}$ between the two tangent circles $\operatorname{Re} K_t(z) = a$ and $\operatorname{Re} K_t(z) = b$ (both tangent at τ).

Horizontal lines in the image and their preimages

Fix $b \in \mathbb{R}$. The horizontal line

$$H_b = \{\, w : \ \operatorname{Im} w = b \,\}$$

pulls back to $\operatorname{Im} K_t(z) = b$. For the Cayley transform,

$$\operatorname{Im} C(u) = \frac{2\ \operatorname{Im} u}{|1-u|^2} = b \iff (1-\operatorname{Re} u)^2 + \Big(\operatorname{Im} u - \frac{1}{b}\Big)^2 = \Big(\frac{1}{b}\Big)^2.$$

Thus for $b \neq 0$,

$$\boxed{\{\operatorname{Im} C(u) = b\} \text{ is a circle tangent to } \partial\mathbb{D} \text{ at } u = 1, \text{ center } (1, 1/b), \text{ radius } 1/|b|.}$$

Rotating back by e^{it} shows that $\{\operatorname{Im} K_t(z) = b\}$ is also a circle tangent to $\partial\mathbb{D}$ at $\tau = e^{it}$. The two families

$$\{\operatorname{Re} K_t = c\}_{c>0}, \qquad \{\operatorname{Im} K_t = b\}_{b\in\mathbb{R}}$$

are orthogonal (conformality).

Horizontal bands. The strip $\beta < \operatorname{Im} w < \delta$ pulls back to the region between the circles $\operatorname{Im} K_t(z) = \beta$ and $\operatorname{Im} K_t(z) = \delta$ (both tangent at τ).

Canonical correspondences

- $z = 0 \mapsto w = 1$.
- $z = \tau = e^{it} \mapsto w = \infty$ (simple pole).
- $\partial\mathbb{D} \setminus \{\tau\} \mapsto i\mathbb{R}$.
- Vertical line $\operatorname{Re} w = c \iff$ circle $\operatorname{Re} K_t(z) = c$ tangent at τ.
- Horizontal line $\operatorname{Im} w = b \iff$ circle $\operatorname{Im} K_t(z) = b$ tangent at τ.
- Vertical/horizontal *bands* $\iff$ regions between pairs of tangent circles.

Harmonic meaning

$$\boxed{\operatorname{Re} K_t(z) = \frac{1-|z|^2}{|\,e^{it} - z\,|^2} = P(z, e^{it})}$$

is exactly the Poisson kernel of the unit disk with pole at the boundary point e^{it}. Hence the circles $\operatorname{Re} K_t(z) = c$ are level sets of harmonic measure; $\operatorname{Im} K_t$ is (up to an additive constant in t) the harmonic conjugate, whose level sets form the orthogonal tangent pencil.

Setup

Let $\tau = e^{it}$ and note

$$K_t(z) = C\big(e^{-it}z\big), \qquad C(\zeta) = \frac{1+\zeta}{1-\zeta}, \qquad z = \frac{w-1}{w+1}.$$

"Vertical" and "horizontal" in the domain are taken *relative to* τ:

$$\text{vertical: } \Re\big(e^{-it}z\big) = a, \qquad \text{horizontal: } \Im\big(e^{-it}z\big) = b.$$

Images of vertical domain lines and bands

Impose $\Re z = a$ with $z = (w-1)/(w+1)$. Writing $w = u + iv$,

$$\Re\Big(\frac{w-1}{w+1}\Big) = a \iff \frac{|w|^2-1}{|w+1|^2} = a \iff \big(u - \tfrac{a}{1-a}\big)^2 + v^2 = \Big(\tfrac{1}{1-a}\Big)^2.$$

Thus

$$\boxed{\Re\big(e^{-it}z\big) = a \implies K_t(z) \text{ lies on the circle } \big(u - \tfrac{a}{1-a}\big)^2 + v^2 = \big(\tfrac{1}{1-a}\big)^2.}$$

In particular:

- $a = 0$ maps to the unit circle $|w| = 1$.
- $a \to 1^-$ produces circles whose limits are vertical lines in the right half-plane.
- The band $a < \Re(e^{-it}z) < b$ maps to the region between the two circles with centers $\frac{a}{1-a}$ and $\frac{b}{1-b}$ on the real axis and radii $1/|1-a|$, $1/|1-b|$.

Images of horizontal domain lines and bands

Impose $\Im z = b$ with $z = (w-1)/(w+1)$. Then

$$\Im\Big(\frac{w-1}{w+1}\Big) = b \iff \frac{2\,\Im w}{|w+1|^2} = b \iff (u+1)^2 + \Big(v - \tfrac{1}{b}\Big)^2 = \Big(\tfrac{1}{b}\Big)^2 \quad (b \neq 0).$$

Therefore

$$\boxed{\Im\big(e^{-it}z\big) = b \implies K_t(z) \text{ lies on the circle } (u+1)^2 + \big(v - \tfrac{1}{b}\big)^2 = \big(\tfrac{1}{b}\big)^2.}$$

Special cases:

- $b = 0$ maps to the positive real axis $\{w > 0\}$.
- For $b \neq 0$ the circle is tangent to the real axis at $w = -1$, with center $(-1, 1/b)$ and radius $1/|b|$.
- The band $\beta < \Im(e^{-it}z) < \delta$ maps to the region between the two tangent circles corresponding to $b = \beta$ and $b = \delta$.

Orthogonality and Möbius geometry

The families

$$\{\Re(e^{-it}z) = a\}_{a\in\mathbb{R}} \quad \text{and} \quad \{\Im(e^{-it}z) = b\}_{b\in\mathbb{R}}$$

are orthogonal in the z-plane, and K_t (being holomorphic/Möbius where defined) preserves angles. Hence their images are two orthogonal pencils of circles/lines in the w-plane.

Band dictionary (domain $\leftrightarrow$ image)

Domain band		Image region
$a < \Re(e^{-it}z) < b$	$\Longleftrightarrow$	between two circles centered on the real axis
$\beta < \Im(e^{-it}z) < \delta$	$\Longleftrightarrow$	between two circles tangent at $w = -1$

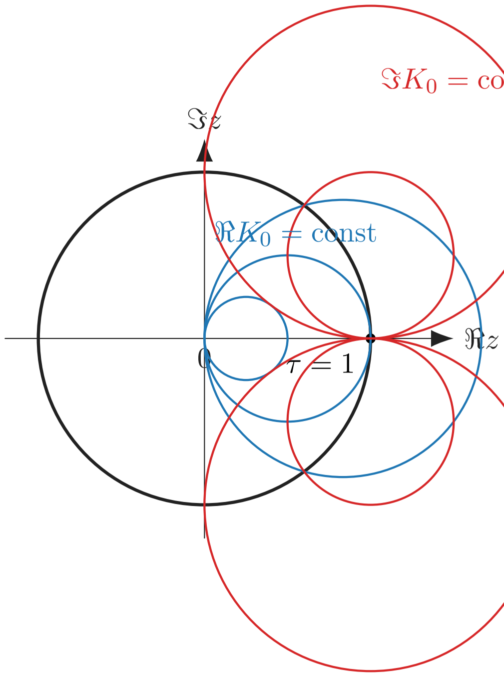

Figure 1: Level sets of $\Re K_0$ (blue) and $\Im K_0$ (red): tangent at 1, orthogonal elsewhere.

Poisson integral on the unit disk. If U is harmonic in $\mathbb{D}$ and has (nontangential) boundary values $g \in C(\partial\mathbb{D})$, then for $z = re^{i\phi} \in \mathbb{D}$,

$$U(re^{i\phi}) = \frac{1}{2\pi}\int_0^{2\pi} g(e^{it})\,P_r(\phi - t)\,dt, \qquad P_r(\theta) = \frac{1-r^2}{1-2r\cos\theta + r^2} = \frac{1-r^2}{|e^{i\theta} - r|^2}.$$

Here $P_r(\theta) \geq 0$ and $\frac{1}{2\pi}\int_0^{2\pi} P_r(\theta)\,d\theta = 1$, so the value at $re^{i\phi}$ is a weighted average of the boundary data on $\partial\mathbb{D}$. Moreover, $P_r(\phi - t) = \Re\left(\frac{e^{it}+re^{i\phi}}{e^{it}-re^{i\phi}}\right)$ and $U(0) = \frac{1}{2\pi}\int_0^{2\pi} g(e^{it})\,dt$.

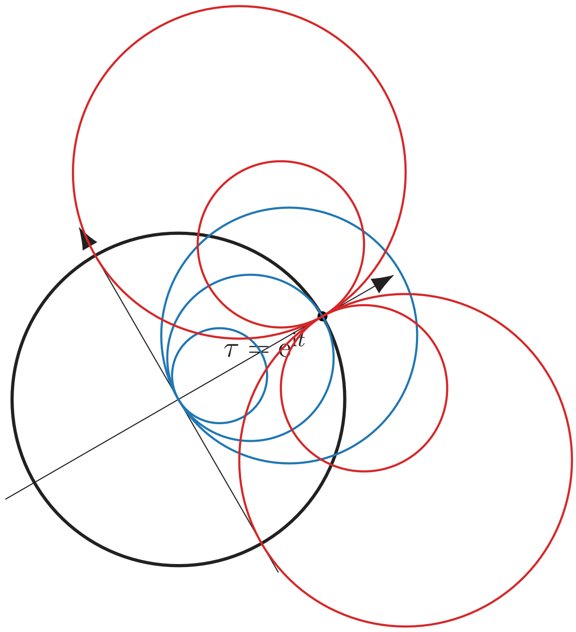

Figure 2: Same grids for K_t, obtained by rotating the $t = 0$ picture by angle t.

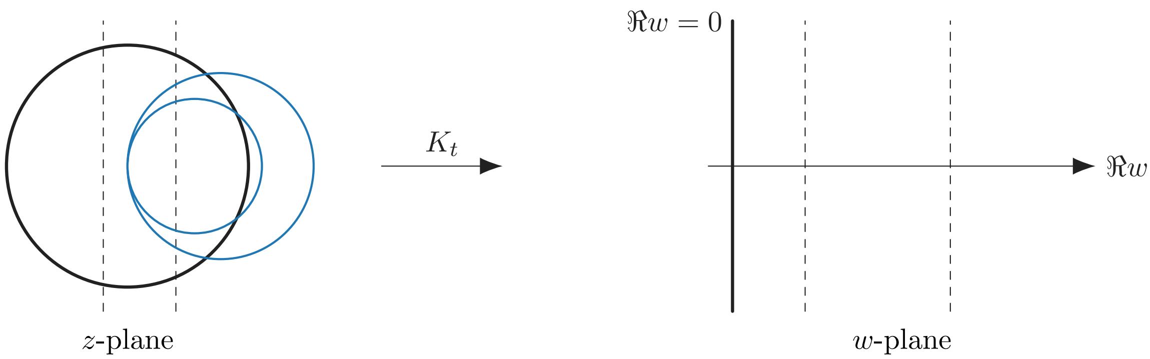

Figure 3: Domain band (between two vertical lines in z) $\mapsto$ vertical strip (between two lines) in the right half-plane.